

Product Specification Document

Service Mobility Dashboard

Full-Stack Final Project

Product Overview

Service Mobility Dashboard is a web-based decision-support system designed for organizations that operate field-service vehicles and manage service tasks in an existing enterprise system of record, such as an ERP platform (e.g., SAP).

The product provides service dispatchers with a unified geographical view of scheduled service tasks, optimized driving routes for each service vehicle, and decision-support tools for evaluating where new service tasks should be assigned.

The system is not intended to replace the organization's existing ERP, operational database, or other core system of record. Instead, it serves as a lightweight operational layer that transforms existing task data into a clear, visual, and actionable planning tool.

Source-System Independence

The original business case arose from an organization using SAP, but the dashboard itself is not tied to SAP-specific APIs or workflows. At the application layer, it consumes a normalized service-task dataset in Supabase. The same product can therefore operate with another ERP, operational database, or enterprise data source when an integration process maps that source data into the expected `service_tasks` schema. Building and operating that source-system connector is outside the scope of the current project.

What Problem Does the Product Solve?

Currently, planning the daily work of field-service vehicles is often performed manually and across several disconnected tools.

Service dispatchers often move between operational data stored in a core enterprise system and external tools such as calendars and mapping applications in order to understand where tasks are located, which vehicle is assigned to each task, and how the daily schedule should be organized.

Although the organization's core operational system contains the required task information, the raw data alone does not provide dispatchers with a clear geographical overview of the day's activity.

This working method creates several problems:

- **Time-consuming planning:** Dispatchers spend valuable time manually combining information from different systems in order to build a daily operational picture.
- **Lack of geographical visibility:** It is difficult to quickly understand how service tasks are geographically distributed between vehicles.
- **Inefficient driving routes:** Vehicles may travel unnecessary distances, revisit the same areas, or perform tasks in an inefficient geographical order.
- **Difficulty assigning new service calls:** When a new task arrives, the dispatcher must manually determine which vehicle and which upcoming workday would be the most geographically suitable assignment.
- **Slow decision-making:** Without a unified operational view, making quick scheduling decisions requires repeatedly checking several systems.
- **Information gaps and human errors:** Working in parallel with the core operational system, calendars, and maps increases the chance of missed information, duplicated work, or inconsistent planning.

The Service Mobility Dashboard addresses these problems by transforming the organization's service-task data into a centralized visual planning environment.

Instead of manually reconstructing the operational picture, the dispatcher can immediately see the relevant service tasks, their locations, their assigned vehicles, and optimized routes on one interactive map.

When a new service request is received, the dispatcher can search for its address and receive geographically relevant vehicle-and-date recommendations for the upcoming days. This allows the dispatcher to make a faster and more informed assignment decision.

Who Are the Users of the Product?

Primary Users - Service Dispatchers

The primary users are service dispatchers responsible for planning and coordinating the daily activity of field-service vehicles.

Their responsibilities include:

- Reviewing scheduled service calls.
- Understanding the geographical distribution of daily tasks.
- Coordinating work between different service vehicles.
- Evaluating where new service calls should be assigned.
- Monitoring customer addresses, time windows, and operational information.
- Making fast scheduling decisions throughout the workday.

The dashboard is designed primarily around the dispatcher's workflow and aims to reduce the number of systems and manual steps required to perform these tasks.

Secondary Users - Operations Managers

Operations managers may also use the system to obtain a high-level overview of field activity and understand how vehicles and service calls are distributed.

Indirect Stakeholders

Field technicians and end customers benefit from the decisions made through the system, although they are not necessarily direct users of the dashboard.

Technicians benefit from more efficient daily routes, while customers may benefit from improved scheduling and better adherence to service windows.

Who Is the Client?

The client is a medium-to-large organization that operates field-service teams or vehicle fleets and already manages its core service operations through an ERP, operational database, or other enterprise system of record.

Examples include:

- Telecommunications companies.
- Maintenance and repair organizations.
- Infrastructure companies.
- Field technician organizations.
- Companies providing on-site installation services.
- Organizations performing complex field deliveries or service visits.

These organizations already possess the operational data required for planning, but need a lightweight visual and analytical layer that makes the information easier for dispatchers to use in day-to-day decision-making.

The organization itself is the client of the product. The end customers receiving the field service are not direct users of the system.

What Are the Business Goals of the Product?

1. Reduce Daily Planning Time

Reduce the amount of time dispatchers spend collecting information from different systems and manually planning the daily activity of service vehicles. The dispatcher should be able to obtain the relevant operational picture from a single interface.

2. Improve Assignment Decisions for New Service Tasks

Help dispatchers decide where a newly received service task should be scheduled. Instead of manually examining several vehicles and future schedules, the system identifies existing service tasks located near the new address and presents possible vehicle-and-date combinations.

This provides the dispatcher with useful geographical information before making the final scheduling decision.

3. Reduce Unnecessary Travel

Improve the geographical order of service stops and reduce unnecessary driving between customer locations.

More efficient routes can contribute to:

- Reduced fuel consumption.
- Reduced vehicle wear.
- Reduced travel time.
- Fewer unnecessary trips between geographical areas.

4. Increase Operational Productivity

Reducing travel and planning overhead allows service teams to spend a larger portion of the workday performing actual service tasks. Over time, this can help the organization make better use of its existing fleet and workforce.

5. Create a Unified Operational Picture

Provide dispatchers with one clear geographical view that combines information that would otherwise need to be reconstructed manually from the organization's core operational system and external tools. This reduces dependence on parallel planning tools and lowers the risk of information gaps and human errors.

6. Improve Service Quality

Better planning and reduced travel time can help the organization adhere more effectively to customer service windows and respond more efficiently to new service requests.

7. Preserve the Existing Core System as the Organizational Source of Truth

The dashboard is designed as a decision-support and visualization layer rather than a replacement for the organization's existing system of record. Service-task data is treated as read-only inside the dashboard, preventing conflicting versions of operational information and keeping the source ERP or core operational system authoritative for service-task management.

What Software Capabilities Are Required to Support the Business Goals?

To achieve the business goals, the product requires the following capabilities:

Secure Authentication

Only authorized organizational users should be able to access operational service information. The application therefore requires secure login and logout functionality.

Daily Task Retrieval

The system must retrieve the service tasks scheduled for a selected date and present only the relevant information to the dispatcher.

Interactive Geographic Visualization

Service tasks must be displayed on an interactive map according to their geographical coordinates. Tasks belonging to different vehicles should be visually distinguishable.

Vehicle-Based Task Grouping and Focus Controls

Tasks must be grouped according to their assigned service vehicle. The dispatcher can focus on or hide individual vehicles so that dense workdays remain easy to inspect without reloading the underlying task data.

Route Optimization

For vehicles with multiple service locations, the system should calculate and display an efficient driving order between the stops.

Date Navigation

Dispatchers must be able to move between different workdays and review scheduled service activity for past or future dates.

Detailed Task Information

The dispatcher must be able to select a service stop and immediately view relevant operational details, such as:

- Customer address.
- Customer name.
- Customer phone number.
- Assigned vehicle.
- Assigned installer.
- Service time window.
- Operational notes.

This reduces the need to return to the core operational system simply to retrieve basic information while planning.

Software Capabilities - Continued

New Address Search

The system must allow the dispatcher to search for the address of a potential new service task, choose from address suggestions, and display the selected geographical location on the map. Address suggestions are localized to Israel and presented for a Hebrew operational workflow.

Recent Address Searches

To reduce repeated typing during day-to-day planning, the search interface keeps a short list of recent address selections on the current browser and allows the dispatcher to reuse them quickly.

Vehicle and Date Recommendations

After selecting the address of a new task, the system examines scheduled tasks during the upcoming four days.

It identifies vehicle-and-date combinations that already contain a service stop within a 20 km radius of the new address. For each vehicle and date, only the closest existing stop is used as the relevant candidate.

Recommendations are ordered by geographical distance and show practical context such as the proposed workday, installer, time window, and distance from the new location.

The recommendation is a decision-support tool. The dispatcher remains responsible for making the final scheduling decision.

Hebrew Right-to-Left Operational Interface

The current product interface is designed in Hebrew and uses a right-to-left layout to match the intended operational environment. Values that are naturally read left-to-right, such as phone numbers and vehicle identifiers, remain visually readable.

Graceful Error Handling

Failures in an external routing service should not prevent the dispatcher from accessing the underlying service-task information.

If route optimization is unavailable for a vehicle, its service stops should still remain visible on the map so that the dispatcher can continue working.

What Are the Main Processes the Product Allows Users to Perform?

1. Secure Login

An authorized employee enters the application and signs in using organizational credentials. After successful authentication, the employee receives access to the operational dashboard. There is no public self-registration flow because the system contains internal organizational information and is intended only for authorized employees.

2. Select a Workday

The dispatcher can select a specific date using the dashboard's date navigation. The system loads the service tasks scheduled for the selected day, allowing the dispatcher to review today's activity as well as other planned workdays.

3. View and Focus the Daily Operational Map

The dispatcher sees all service tasks scheduled for the selected date on a centralized interactive map. Tasks are grouped by their assigned vehicle, allowing the dispatcher to understand which vehicles are active, which stops belong to each vehicle, where activity is geographically concentrated, and how routes relate to one another.

The dispatcher can focus on an individual vehicle or temporarily hide a vehicle to reduce visual clutter during busy workdays.

4. Review an Optimized Route for Each Vehicle

For each vehicle with multiple valid service locations, the system calculates an optimized driving route between its stops. The resulting route is displayed directly on the map together with numbered service-stop markers, providing an immediate visual understanding of the suggested order of travel.

5. Inspect a Specific Service Task

The dispatcher can select a stop either from the map or from the vehicle's task list. The map focuses on the selected location and displays relevant service information without requiring the dispatcher to leave the dashboard.

6. Search for a New Service Address

When a new service request is received, the dispatcher can enter its address into the dashboard and choose from suggested Israeli addresses. The selected location is displayed on the map. Recent address selections are also available for quick reuse on the same browser.

Main Processes - Continued

7. Receive Scheduling Recommendations

After a new address has been selected, the system examines service tasks already scheduled during the next four days and searches for vehicles that already have a stop within 20 km of the new location.

The dispatcher receives a list of candidate assignments containing information such as:

- Proposed workday.
- Distance from the new task to a nearby existing stop.
- Assigned installer.
- Existing service time window.

The recommendations are sorted by distance so that the most geographically relevant options appear first.

8. Compare a Recommendation With the Existing Schedule

The dispatcher can select a recommendation and move directly to that candidate's workday while keeping the searched location available on the map. This makes it possible to compare the proposed location with the vehicle's existing stops and route before deciding whether the assignment is operationally appropriate.

The system assists the decision but does not automatically modify the schedule in the upstream system of record.

9. Logout

When the dispatcher finishes working with the system, they can securely log out and end the authenticated session.

Product Scope and Boundaries

The current product is designed primarily as a visualization and decision-support layer for service dispatchers.

The system:

- Reads normalized service-task information originating from an enterprise source system (e.g., SAP) through the Supabase mirror.
- Displays and organizes that information geographically.
- Calculates optimized routes for daily service activity.
- Helps dispatchers evaluate potential assignments for new service requests.
- Supports rapid operational comparison using date navigation, vehicle focus controls, and recent address searches.

The system does not currently:

- Replace the organization's existing ERP, operational database, or other system of record.
- Directly modify service-task records in the upstream ERP or core operational system.
- Automatically assign a new task to a vehicle.
- Automatically change the operational schedule based on a recommendation.
- Perform full workforce scheduling optimization based on technician skills, workload, capacity, or time-window conflicts; current recommendations are primarily based on geographical proximity within the upcoming four-day window.

These boundaries are intentional. The goal of the current version is to provide dispatchers with better information and faster decision-support while keeping the organization's existing source system authoritative for operational data.